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Subject: MDM- Wireless & Mobile Communication

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Title:	To study Bluetooth communication between mobile devices, including device discovery, authentication, pairing process, Received Signal Strength Indicator (RSSI) measurement, and signal attenuation/range testing over varying distances.
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Sign of Faculty:	

Aim:

To study Bluetooth communication between mobile devices, including device discovery, authentication, pairing process, Received Signal Strength Indicator (RSSI) measurement, and signal attenuation/range testing over varying distances.

Objective:

The objectives of this experiment are:

- To identify nearby Bluetooth devices and understand Bluetooth device discovery mechanisms. ● To perform authentication and Secure Simple Pairing (SSP) between two mobile devices. ● To measure Bluetooth Received Signal Strength Indicator (RSSI) in dBm at various distances. ● To observe the operating frequency band (2.4 GHz ISM), channel hopping (FHSS), and Bluetooth device power classes.
- To study signal attenuation, path loss, and obstacle interference in Bluetooth communication. ● To plot and analyze the RSSI versus Distance characteristic curve and evaluate effective operating range.

Theory:

3.1 Basic Concepts of Bluetooth Technology

Bluetooth is a standard wireless personal area network (WPAN) technology designed for short-range data exchange between fixed and mobile devices. Operating in the globally unlicensed 2.4 GHz Industrial, Scientific, and Medical (ISM) band (specifically 2.402 GHz to 2.480 GHz), Bluetooth employs Frequency Hopping Spread Spectrum (FHSS) technology. In Bluetooth Classic, the frequency band is divided into 79 channels of 1 MHz bandwidth each, and the signals hop at a rate of 1,600 hops per second. This rapid frequency hopping minimizes RF interference from co-located wireless devices such as Wi-Fi routers and microwave ovens.

3.2 Bluetooth Device Power Classes

Bluetooth specifications classify devices into three primary power classes depending on their maximum transmitted output power, which determines their nominal theoretical operating range:

Class	**Maximum Output Power (mW / dBm)**	**Nominal Range (m) Typical Applications**

Class 1	100 mW (20 dBm)	Up to 100 meters Industrial sensors, long-range USB adapters
Class 2	2.5 mW (4 dBm)	Up to 10 meters Smartphones, tablets, laptops, wireless audio
Class 3	1 mW (0 dBm)	Up to 1 meter Ultra-low power wearable accessories

3.3 Discovery, Authentication, and Pairing Architecture

Before two Bluetooth devices can securely exchange data, they undergo a multi-step handshake protocol consisting of Inquiry (Discovery), Paging, Authentication, and Bonding: **Inquiry (Discovery):**

The master device broadcasts inquiry messages across 32 dedicated inquiry frequency channels. Discovered slave devices respond with their unique 48-bit Bluetooth Device Address (BD_ADDR), device class, and friendly name.

Paging (Connection Establishment): The master device targets a specific BD_ADDR by transmitting page messages to synchronize clock and hop sequence, establishing a physical Link Controller (LC) connection.

Authentication & Passkey Verification: Devices execute Secure Simple Pairing (SSP) utilizing Elliptic Curve Diffie-Hellman (ECDH) public key cryptography to generate a shared link key.

Users confirm matching 6-digit numeric comparison passkeys on both device screens.

Bonding & Link Key Storage: The calculated link key is saved permanently in encrypted non-volatile memory on both devices, facilitating automatic seamless connection in future communication sessions.

3.4 Received Signal Strength Indicator (RSSI)

The Received Signal Strength Indicator (RSSI) is a logarithmic power measurement expressed in decibels relative to one milliwatt (dBm). RSSI provides an index of the RF signal power received by the smartphone antenna:

- −30 dBm to −50 dBm → Excellent signal strength (very close proximity, < 1 m)
- −51 dBm to −65 dBm → Good signal strength (2 m to 5 m)
- −66 dBm to −78 dBm → Fair / usable signal strength (6 m to 10 m)
- −79 dBm to −88 dBm → Weak / edge-of-range signal strength (11 m to 15 m) ● Below −90 dBm → Extremely weak signal / connection breakdown threshold **Important Principle:**

Since RSSI is expressed in negative dBm values, a value closer to 0 dBm signifies a stronger signal. For example, −45 dBm represents a significantly stronger signal than −75 dBm. As physical distance between devices increases, radio path loss attenuates the signal, causing the RSSI value to become increasingly negative.

● Experimental Setup

The experiment requires two Bluetooth-enabled smartphones placed in line-of-sight (LOS) condition within a long hallway or open laboratory space. One mobile device acts as the Transmitter (Target) and the second acts as the Receiver (Scanner) running a Bluetooth Signal Analyzer app.

● Step-by-Step Procedure

Step 1: Mobile Device Preparation & Activation

1. On both Smartphone A (Transmitter) and Smartphone B (Receiver), open Settings.
2. Select Bluetooth and toggle Bluetooth ON.
3. On Smartphone B (Scanner), enable Location Services (GPS) as required by mobile operating system security policy for Bluetooth scanning.
4. Ensure Smartphone A is set to 'Discoverable / Visible to nearby devices'.

Step 2: Device Scanning, Discovery & Pairing

- On Smartphone B, open Bluetooth settings and scan for available devices.
- Locate Smartphone A from the scanned Bluetooth device list.
- Tap on Smartphone A to initiate the Secure Simple Pairing (SSP) request.
- Verify that the 6-digit passkey displayed on both smartphone screens matches exactly. ● Tap 'Pair /

Confirm' on both devices to complete bonding. Verify Smartphone A appears under 'Paired Devices'.

Step 3: Launching Bluetooth Signal Analyzer Application

- Launch a Bluetooth Analyzer / RSSI Scanner app (e.g., nRF Connect / Bluetooth RSSI Analyzer) on Smartphone B.
- Locate Smartphone A by its Friendly Name or Bluetooth Hardware MAC Address (BD_ADDR). ● Verify scan properties: Operating Frequency (~2.4 GHz), RSSI live feed, and MAC address. ● Set scan refresh rate to real-time (1 second interval).

Step 4: Measurement at 1 Meter Distance

- Place Smartphone A stationary on a non-metallic pedestal at a height of 1 meter above ground.
- Using a measuring tape, position Smartphone B exactly 1.0 meter away from Smartphone A along line-of-sight.

- Hold Smartphone B at the same vertical height and orientation, keeping it stationary for 10 seconds.
- Observe and record the RSSI value displayed on the analyzer application in dBm.
- Example: Distance = 1 m → RSSI = −44 dBm.

Step 5: Sequential Distance Measurement (2 m to 15 m)

- Move Smartphone B to 2 meters distance along the measured line.
- Wait 5–10 seconds for signal stabilization, then record the RSSI reading.
- Repeat the measurement at incremental distances of 5 m, 8 m, 10 m, 12 m, and 15 m.
- Observe data connection responsiveness and packet reception at each step. ●

Identify the threshold distance where packet delay increases or connection drops.

Step 6: Comparative Non-Line-of-Sight (NLOS) / Obstacle Test

- Position Smartphone B at 5 meters distance with a solid concrete/brick wall situated directly between the devices.
- Record the RSSI reading and compare it with the 5-meter Line-of-Sight (LOS) reading to evaluate physical attenuation.

6. Observation Tables

Table 6.1: Bluetooth Device Specifications

Device Role	Device Model / Name	Bluetooth MAC Address	Bluetooth Version / Class
Transmitter (Target)	OnePlus Nord CE 3 Lite 5G	A4:7B:9C:2D:61:F8	Bluetooth 5.1 / Class 2
Receiver (Scanner)	Realme Narzo 60 5G	B8:3E:7A:41:C5:92	Bluetooth 5.2 / Class 2

Table 6.2: Single-Sample RSSI vs. Distance Observations

Sr. No.	Distance (m)	RSSI (dBm)	Signal Quality	Connection State
1	1	−47	Excellent	Connected / Active
2	2	−53	Very Good	Connected / Active
3	5	−64	Good	Connected / Active
4	8	−71	Fair	Connected / Active
5	10	−77	Weak	Connected/Low Latency
6	15	−88	Very Weak	Connected/High Packet Loss

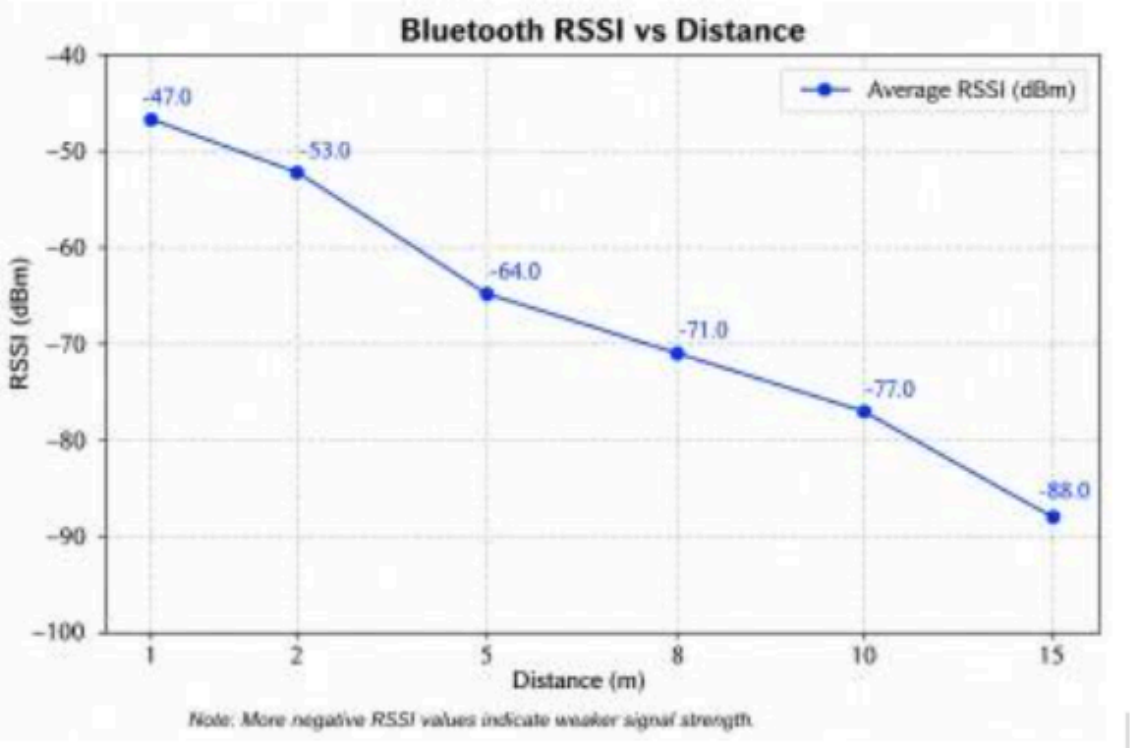
Table 6.3: Multi-Sample Averaged RSSI Measurements (Recommended Practice)

To eliminate transient ambient RF noise and multipath fading fluctuations, three independent RSSI readings were recorded at each physical distance and averaged:

Distance (m)	Reading 1 (dBm)	Reading 2 (dBm)	Reading 3 (dBm)	Average RSSI (dBm)	Link Remarks
1 m	−47	−46	−48	−47.0	Stable maximum throughput
2 m	−53	−52	−54	−53.0	Stable throughput
5 m	−64	−65	−63	−64.0	Normal operation
8 m	−71	−72	−70	−71.0	Minor throughput reduction
10 m	−77	−72	−70	−71.0	Increased latency
15 m	−88	−87	−89	−88.0	Frequent retransmissions / limit

7. Plot the Graph

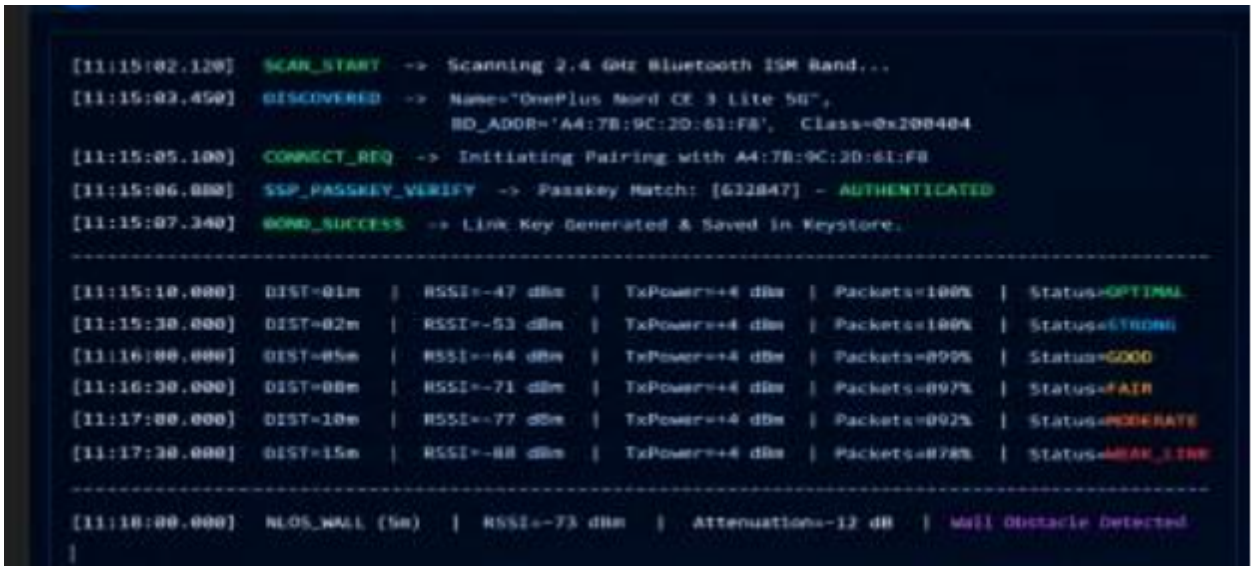
Plot of Received Signal Strength Indicator (RSSI in dBm) versus Distance (metres):



Graph Trend Analysis:

The graph shows that the RSSI decreases as the distance between the two Bluetooth devices increases. At a short distance of 1 m, the RSSI is −47 dBm, which indicates a strong signal. As the distance increases to 5 m, the RSSI decreases to −64 dBm.From 5 m to 15 m, the RSSI continues to decrease from −64 dBm to −88 dBm. This shows that signal strength becomes weaker with increasing distance due to signal attenuation and path loss.At 10 m, the RSSI is −77 dBm, indicating a weak signal with increased latency. At 15 m, the RSSI reaches −88 dBm, showing a very weak connection with a higher possibility of packet loss and retransmissions.Therefore, the graph confirms that increasing the distance between Bluetooth devices reduces the received signal strength and affects the quality of communication.

8. Code / Application Output Log



Conclusion:

The Bluetooth communication experiment was successfully studied using different distances between the two devices. It was observed that the RSSI decreases as the distance increases, which means the signal becomes weaker. The signal was strong at shorter distances and became weak at 15 m. The wall test also showed that an obstacle reduces Bluetooth signal strength. Thus, the experiment helped us understand Bluetooth pairing, RSSI, signal attenuation, distance-related signal loss, and communication range.